



# Marketing

## Functional Data Analysis on Monthly Sales

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STAT 591

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# Introduction

## Objective:

Analyze and understand temporal sales patterns across different U.S. regions and states using Functional Data Analysis (FDA) techniques.

## Goals:

- Transform sales time series data into smooth functional representations
- Apply B-spline smoothing to monthly sales curves
- Perform Functional Principal Component Analysis (FPCA) to extract key variation modes
- Use clustering methods to group states with similar sales trajectories

**Data Source:** (<https://www.kaggle.com/datasets/rohitsahoo/sales-forecasting/data>)

Superstore Sales dataset from Kaggle, containing:

- Order dates
- Sales amounts
- Region and state information

# Original Dataset

Total number of orders: 9800

Time range: from 01/2015 to 12/2018

Key variables:

Order Date: Date of transaction

Sales: Sales amount in USD

Region: One of {Central, East, South, West}

State: U.S. state where the sale occurred

| Row ID | Order ID  | Order Date | Ship Date | Ship Mode      | Customer | Customer Segment  | Country   | City          | State           | Postal Code | Region | Product ID | Product Name | Sales          |          |
|--------|-----------|------------|-----------|----------------|----------|-------------------|-----------|---------------|-----------------|-------------|--------|------------|--------------|----------------|----------|
| 1      | CA-2017-1 | #####      | #####     | Second Class   | CG-12520 | Claire Gutierrez  | Consumer  | United States | Henderson       | Kentucky    | 42420  | South      | FUR-BO-1     | Bush Something | 261.96   |
| 2      | CA-2017-1 | #####      | #####     | Second Class   | CG-12520 | Claire Gutierrez  | Consumer  | United States | Henderson       | Kentucky    | 42420  | South      | FUR-CH-1     | Hon Deluxe     | 731.94   |
| 3      | CA-2017-1 | #####      | 16/06/201 | Second Class   | DV-13045 | Darrin Van        | Corporate | United States | Los Angeles     | California  | 90036  | West       | OFF-LA-1C    | Self-Adhesive  | 14.62    |
| 4      | US-2016-1 | #####      | 18/10/201 | Standard Class | SO-20335 | Sean O'Donnell    | Consumer  | United States | Fort Lauderdale | Florida     | 33311  | South      | FUR-TA-1C    | Bretford Case  | 957.5775 |
| 5      | US-2016-1 | #####      | 18/10/201 | Standard Class | SO-20335 | Sean O'Donnell    | Consumer  | United States | Fort Lauderdale | Florida     | 33311  | South      | OFF-ST-10    | Eldon Fold     | 22.368   |
| 6      | CA-2015-1 | 9/6/2015   | 14/06/201 | Standard Class | BH-11710 | Brosina Hernandez | Consumer  | United States | Los Angeles     | California  | 90032  | West       | FUR-FU-1C    | Eldon Express  | 48.86    |
| 7      | CA-2015-1 | 9/6/2015   | 14/06/201 | Standard Class | BH-11710 | Brosina Hernandez | Consumer  | United States | Los Angeles     | California  | 90032  | West       | OFF-AR-1C    | Newell 322     | 7.28     |
| 8      | CA-2015-1 | 9/6/2015   | 14/06/201 | Standard Class | BH-11710 | Brosina Hernandez | Consumer  | United States | Los Angeles     | California  | 90032  | West       | TEC-PH-1C    | Mitel 5320     | 907.152  |
| 9      | CA-2015-1 | 9/6/2015   | 14/06/201 | Standard Class | BH-11710 | Brosina Hernandez | Consumer  | United States | Los Angeles     | California  | 90032  | West       | OFF-BI-10    | DXL Angle      | 18.504   |
| 10     | CA-2015-1 | 9/6/2015   | 14/06/201 | Standard Class | BH-11710 | Brosina Hernandez | Consumer  | United States | Los Angeles     | California  | 90032  | West       | OFF-AP-1C    | Belkin F5C     | 114.9    |

# Data Preprocessing

Prepared **two datasets**: one aggregated by region, the other by state.

For each dataset, monthly sales were **aggregated** for each region or state.

**Filtered** out states with fewer than 5 months of available data to ensure reliability.

Created a **numeric time** variable using year and month, e.g., 2015.000, 2016.667.

|    | Region  | Year | Month | Monthly_Sales | Time     |
|----|---------|------|-------|---------------|----------|
| 1  | Central | 2015 | 1     | 1533.966      | 2015.000 |
| 2  | Central | 2015 | 2     | 1233.174      | 2015.083 |
| 3  | Central | 2015 | 3     | 5827.602      | 2015.167 |
| 4  | Central | 2015 | 4     | 3712.340      | 2015.250 |
| 5  | Central | 2015 | 5     | 4044.522      | 2015.333 |
| 6  | Central | 2015 | 6     | 9374.107      | 2015.417 |
| 7  | Central | 2015 | 7     | 6740.574      | 2015.500 |
| 8  | Central | 2015 | 8     | 3022.183      | 2015.583 |
| 9  | Central | 2015 | 9     | 34254.866     | 2015.667 |
| 10 | Central | 2015 | 10    | 8965.757      | 2015.750 |

|    | State   | Year | Month | Monthly_Sales | Time     |
|----|---------|------|-------|---------------|----------|
| 1  | Alabama | 2015 | 4     | 1224.880      | 2015.250 |
| 2  | Alabama | 2015 | 5     | 180.930       | 2015.333 |
| 3  | Alabama | 2015 | 10    | 2216.800      | 2015.750 |
| 4  | Alabama | 2015 | 11    | 14.670        | 2015.833 |
| 5  | Alabama | 2015 | 12    | 2501.810      | 2015.917 |
| 6  | Alabama | 2016 | 3     | 518.490       | 2016.167 |
| 7  | Alabama | 2016 | 6     | 976.290       | 2016.417 |
| 8  | Alabama | 2016 | 7     | 457.040       | 2016.500 |
| 9  | Alabama | 2016 | 9     | 1940.150      | 2016.667 |
| 10 | Alabama | 2017 | 5     | 1007.680      | 2017.333 |

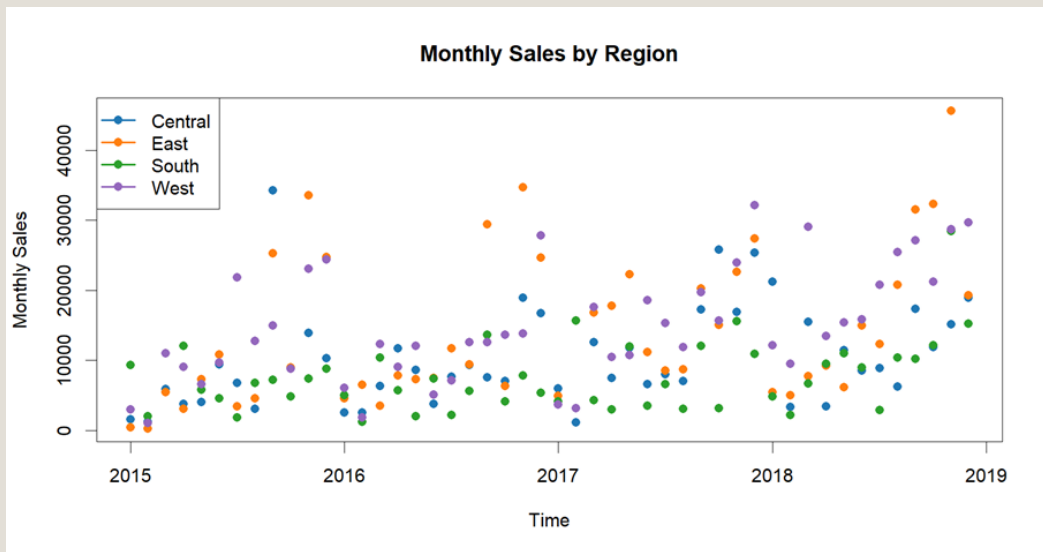
# Regional Sales Patterns

Sales data plotted over time for each of the four regions

Noticeable upward trends, especially after 2017

Clear regional differences in sales levels and variability

Scatter plot of monthly sales by region (2015–2019):



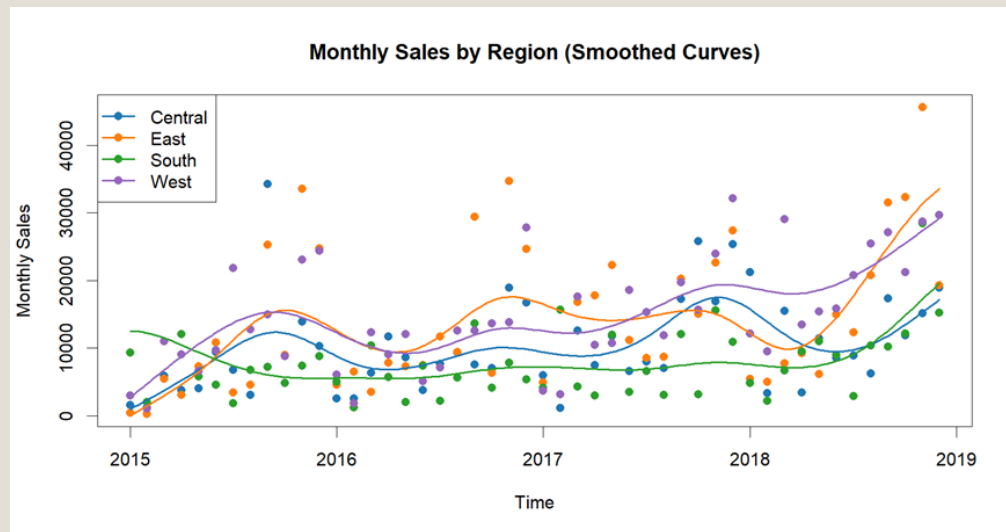
# Smoothing of Regional Sales Trends

Applied B-spline basis smoothing to monthly sales curves for each region.

Used cross-validation to select the optimal number of basis functions for each region ( $n = 14$ ).

Employed a regularization parameter ( $\lambda = 0.8$ ) to control the trade-off between smoothness and fidelity.

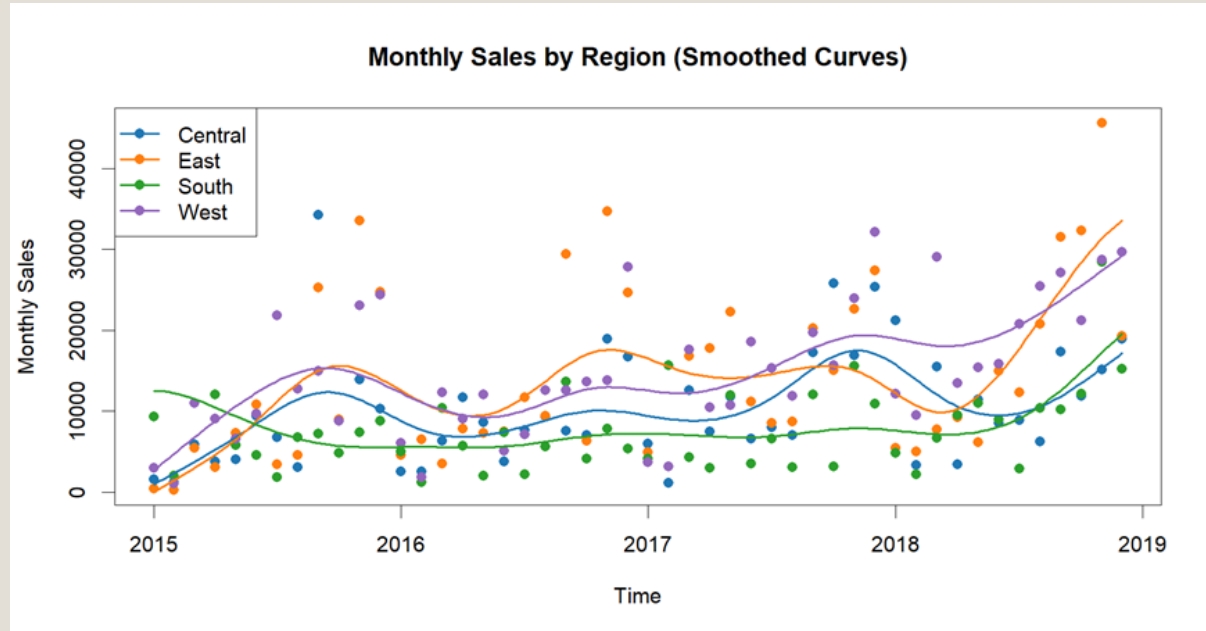
Visualization shows consistent seasonal and structural patterns across regions.



# Smoothing of Regional Sales Trends

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- East and West regions show a consistent upward sales trend, especially after 2017
- Central region shows moderate growth with some seasonal patterns
- South remains relatively flat with lower sales overall



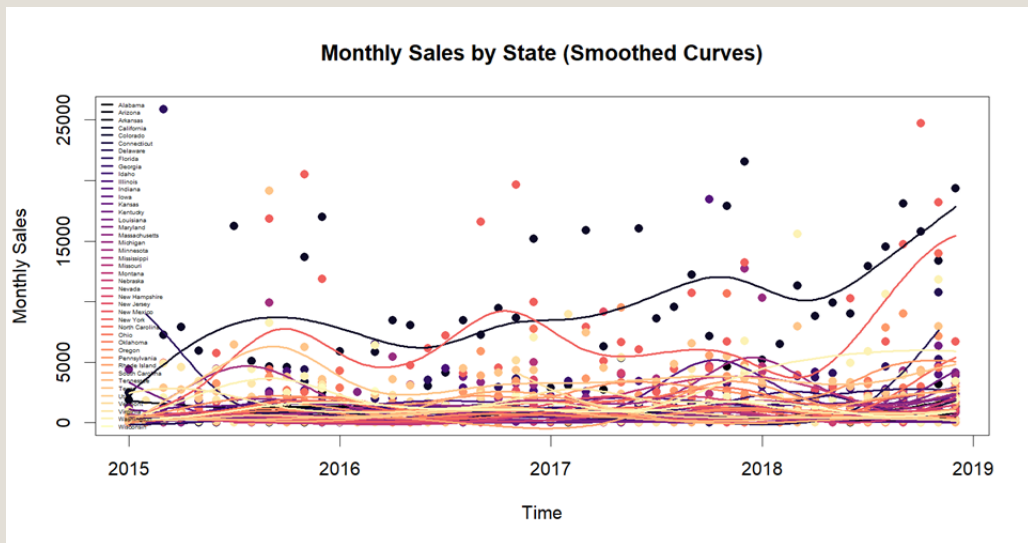
# State-Level Smoothed Sales

Substantial variability in sales trajectories across states.

A few states (e.g., California, New York) exhibit strong upward trends, suggesting robust market growth. Most states show relatively flat or mildly increasing patterns.

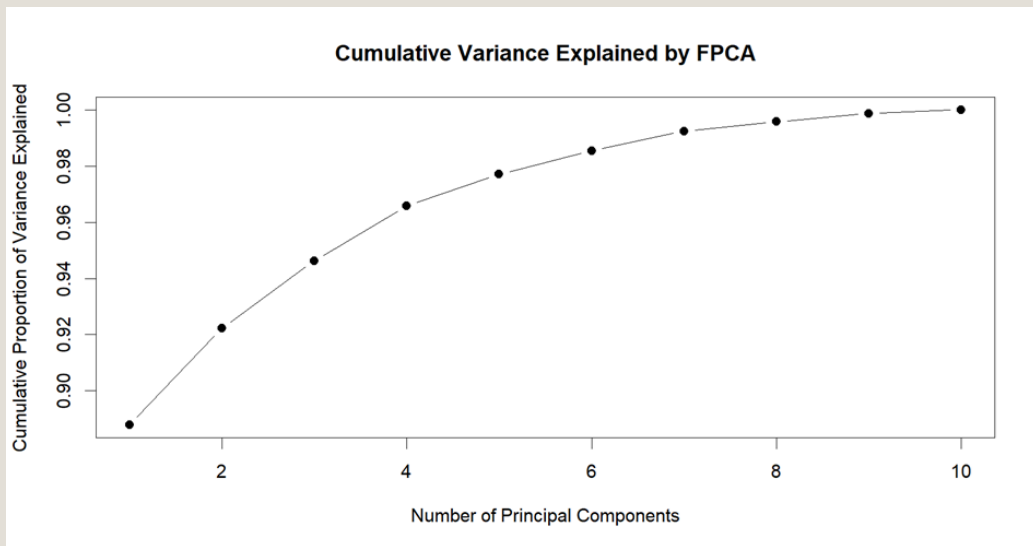
Clear structural patterns visible in higher-sales states, while others remain noisy or stagnant.

Potential regional clusters are hinted at, motivating further functional clustering.



# fPCA

- Applied FPCA to state-level smoothed sales curves
- Used B-spline smoothing to represent each curve in a functional space
- Selected 4 principal components, capturing over 95% of total variance
- Reduced 49-dimensional functional curves to a 4-dimensional representation



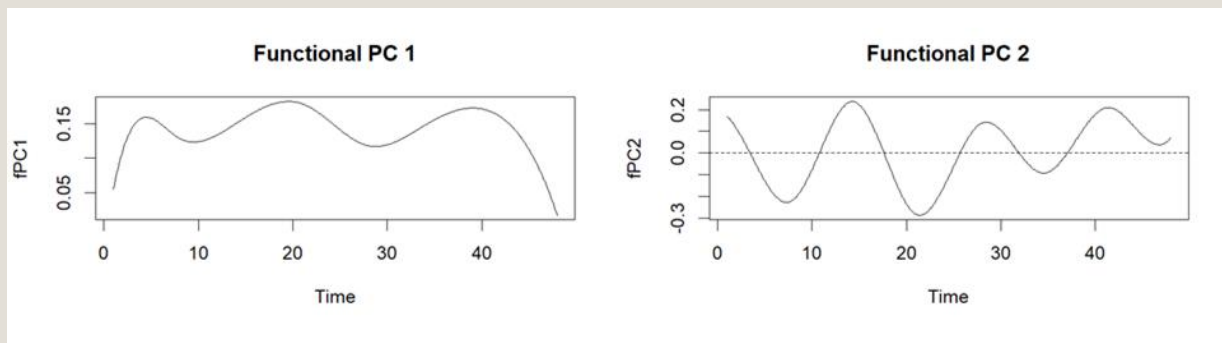
# fPCA

## Functional PC 1

- Pattern: Broadly positive curve with two large humps around time points 10 and 30
- Explanation: fPC1 captures the overall average sales trend

## Functional PC 2

- Pattern: Oscillating curve with alternating positive and negative peaks
- Explanation: fPC2 reflects seasonal or cyclical fluctuations in sales

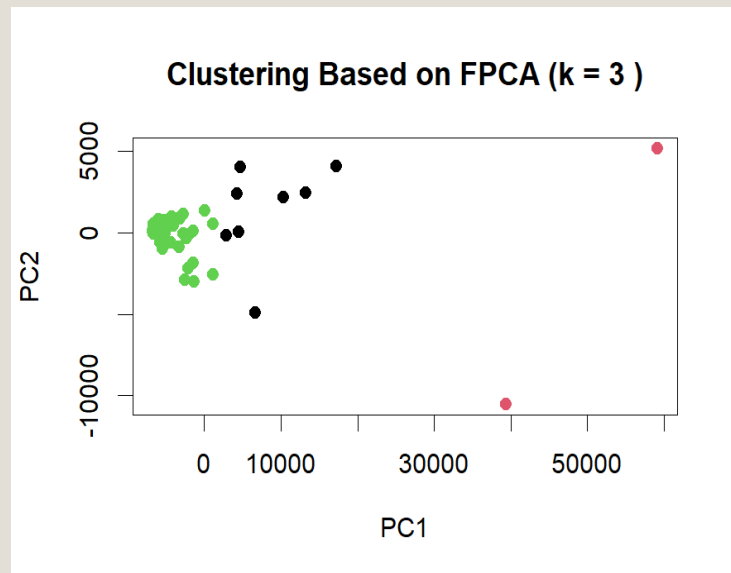
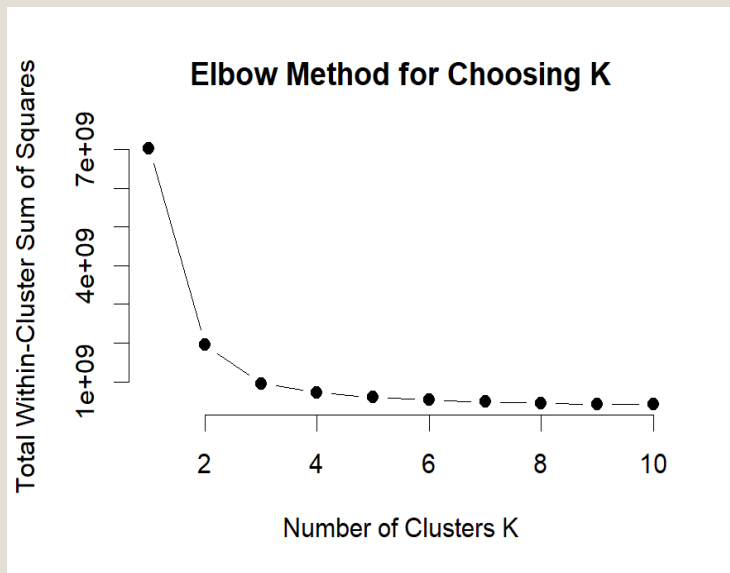


# Clustering States Using fPCA

Identified distinct patterns among states based on functional features using fPCA scores.

Applied K-means clustering on the first 4 fPCA scores and selected  $k = 3$  via the Elbow Method.

Three clusters clearly emerged, showing that FPCA preserves key variation while enabling effective dimensionality reduction.



# Clustering States Using fPCA

## Cluster Summary

- Cluster 1: 9 states (e.g., Florida, Illinois, Michigan)
- Cluster 2: 2 states (California, New York)
- Cluster 3: 38 states (majority of the U.S.)

## Key Insights

- Cluster 3 dominates, indicating a large group of states share similar functional profiles
- Cluster 1 contains diverse states both geographically and demographically, potentially representing a unique usage or policy pattern
- Cluster 2, consisting of California and New York, may reflect distinct large-state behaviors that differ significantly from the rest

| State                | Cluster |
|----------------------|---------|
| Florida              | 1       |
| Illinois             | 1       |
| Michigan             | 1       |
| Ohio                 | 1       |
| Pennsylvania         | 1       |
| Texas                | 1       |
| Virginia             | 1       |
| Washington           | 1       |
| California           | 2       |
| New York             | 2       |
| Alabama              | 3       |
| Arizona              | 3       |
| Arkansas             | 3       |
| Colorado             | 3       |
| Connecticut          | 3       |
| Delaware             | 3       |
| District of Columbia | 3       |
| Georgia              | 3       |

# Thank You

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